

FAIRFIELD, IL, USA

WMO: 744656

Lat: 38.379N

Lon: 88.413W

Elev: 133

StdP: 99.74

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-11.0	-19.6	0.7	-10.7	-17.1	0.8	-8.7	9.2	9.7	8.4	8.5	1.5	300	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.1	24.1	32.9	24.1	32.2	24.0	27.1	31.3	26.2	30.6	25.4	29.8	2.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.8	29.8	24.9	20.3	28.8	24.0	19.1	27.9	86.1	31.4	81.7	31.0	78.4	29.5	32.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	5.9	-17.7	36.1	2.6	1.8	-19.6	37.4	-21.1	38.5	-22.6	39.5	-24.5	40.8		
(5)			-18.2	28.5	2.5	1.6	-20.0	29.6	-21.4	30.6	-22.8	31.5	-24.6	32.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	0.0	1.7	8.0	13.9	19.2	24.0	25.4	24.4	21.2	14.2	7.6	2.6	(6)
(7)		DBStd	10.24	6.41	6.30	6.10	5.14	4.59	3.18	2.74	2.83	3.72	5.20	5.46	5.56	(7)
(8)		HDD10.0	1054	317	241	113	22	1	0	0	0	16	108	237		(8)
(9)		HDD18.3	2537	569	466	324	147	48	3	0	1	16	151	323	488	(9)
(10)		CDD10.0	2357	5	8	51	139	287	419	477	446	335	147	37	7	(10)
(11)		CDD18.3	797	0	0	3	15	76	172	218	188	101	23	1	0	(11)
(12)		CDH23.3	7554	0	0	17	119	635	1702	2188	1751	922	212	7	0	(12)
(13)		CDH26.7	2969	0	0	1	13	186	692	946	723	349	59	1	0	(13)
(14)	Wind	WSAvg	2.3	2.8	2.8	3.0	3.0	2.3	1.9	1.5	1.4	1.6	2.0	2.4	2.5	(14)
(15)	Precipitation	PrecAvg	1123	92	63	92	111	127	102	91	85	77	88	104	82	(15)
(16)		PrecMax	1408	220	125	229	266	274	232	174	224	203	219	230	178	(16)
(17)		PrecMin	926	15	11	24	34	33	28	25	23	3	14	2	43	(17)
(18)		PrecStd	155	47	33	47	62	68	54	42	51	48	54	53	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	20.5	26.0	27.9	32.4	35.9	37.4	36.3	33.9	31.7	24.0	18.3	(19)
(20)			MCWB	15.8	15.8	17.6	18.9	23.1	23.3	24.3	23.6	21.3	19.2	16.4	15.4	(20)
(21)		2%	DB	16.1	17.6	22.6	26.7	30.2	33.5	34.0	34.0	32.4	28.0	21.2	16.1	(21)
(22)			MCWB	13.4	13.6	16.7	18.0	22.0	24.0	25.6	23.8	22.1	19.8	16.2	14.0	(22)
(23)		5%	DB	12.7	14.4	20.2	24.4	28.8	32.4	32.7	32.5	30.9	26.0	18.7	13.3	(23)
(24)			MCWB	10.6	11.3	15.3	17.3	21.2	23.8	25.1	24.3	22.3	18.1	14.5	11.0	(24)
(25)		10%	DB	9.1	11.7	17.6	22.6	27.4	31.0	31.5	31.0	28.6	22.9	16.4	11.1	(25)
(26)			MCWB	7.0	8.6	13.9	16.3	20.7	23.0	24.6	24.2	21.1	17.2	12.3	9.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	16.9	19.6	21.3	25.0	27.3	28.6	28.6	25.6	23.3	19.4	16.6	(27)
(28)			MCD B	18.0	19.1	23.2	26.0	29.6	31.6	33.8	32.8	29.3	28.9	21.5	17.7	(28)
(29)		2%	WB	14.0	14.7	17.9	20.0	23.7	26.2	27.3	26.8	24.4	21.7	17.4	14.4	(29)
(30)			MCD B	15.8	17.0	21.0	24.7	28.1	31.1	31.7	30.6	29.5	26.1	19.1	16.1	(30)
(31)		5%	WB	11.1	12.0	16.2	18.7	22.7	25.3	26.4	25.7	23.5	20.2	15.3	11.8	(31)
(32)			MCD B	12.3	14.2	19.8	22.6	27.0	29.9	30.9	30.0	28.2	23.8	18.1	13.1	(32)
(33)		10%	WB	7.1	8.5	14.3	17.3	21.7	24.3	25.5	24.8	22.6	18.4	12.9	8.9	(33)
(34)			MCD B	8.9	11.2	17.2	20.9	25.8	28.8	29.9	29.0	26.5	21.7	15.6	10.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	8.9	9.9	10.9	10.3	10.9	11.1	12.1	11.5	9.9	8.0	(35)	
(36)			MCD BR	11.6	13.0	12.4	12.8	11.9	12.2	11.7	13.1	13.9	14.0	13.0	10.4	(36)
(37)			MCWBR	10.1	10.3	8.0	7.1	5.4	5.0	5.1	5.7	5.6	6.6	8.1	8.9	(37)
(38)		5% WB	MCD BR	10.6	12.1	11.3	10.5	10.5	10.4	10.1	11.0	11.3	11.5	11.0	9.5	(38)
(39)			MCWBR	10.1	10.7	7.7	6.8	5.6	5.3	5.4	5.8	5.6	6.6	9.0	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.325	0.363	0.402	0.436	0.440	0.466	0.468	0.411	0.360	0.322	0.309	(40)	
(41)		taud	2.455	2.400	2.358	2.253	2.213	2.221	2.156	2.140	2.291	2.430	2.480	2.470	(41)	
(42)		Ebn at Noon	868	901	896	876	849	843	817	805	834	850	848	844	(42)	
(43)		Edh at Noon	82	98	113	134	142	141	149	148	120	95	79	75	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.97	2.74	3.73	4.92	5.54	6.40	6.20	5.63	4.69	3.42	2.30	1.68	(44)	
(45)		RadStd	0.19	0.30	0.35	0.28	0.48	0.41	0.39	0.30	0.32	0.36	0.26	0.16	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78

Nomenclature: See separate page